

CSE4204 — Mobile Computing Lab

Week 04 Submission

System Design & Software Architecture

Smart Farmer Assistant

AI-Based Agriculture Support System

Team Name	CSE4204-8D-T03
Section	8D
Project Title	Smart Farmer Assistant: AI-Based Agriculture Support System
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GitHub Repo	github.com/sakib-foysal/CSE4204-8D-T03-SmartFarmer-Assistant
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ER Diagram

The database is implemented in PostgreSQL with six entities. The users table is the central entity; disease_history, chat_history, and fertilizer_recommendations all carry a user_id foreign key. The weather_data and market_prices tables are populated from external APIs and have no user foreign key.

i. Entities & Attributes

users

Column	Type	Constraint	Description
id	VARCHAR(36)	PRIMARY KEY	UUID — unique user identifier
name	VARCHAR(100)	NOT NULL	Full name of the farmer
email	VARCHAR(150)	UNIQUE, NOT NULL	Login email address
password	VARCHAR(255)	NOT NULL	Bcrypt-hashed password
phone	VARCHAR(20)	NULL	Mobile phone number
role	VARCHAR(10)	DEFAULT 'farmer'	farmer or admin
created_at	TIMESTAMP	DEFAULT NOW()	Account creation timestamp

disease_history

Column	Type	Constraint	Description
id	VARCHAR(36)	PRIMARY KEY	UUID
user_id	VARCHAR(36)	FK → users.id	Owning farmer
image_url	TEXT	NOT NULL	Path/URL of uploaded crop image
prediction	VARCHAR(150)	NOT NULL	AI-predicted disease name
confidence	DECIMAL(5,2)	NOT NULL	Confidence score (0–100)
treatment	TEXT	NULL	AI-generated treatment advice
date	TIMESTAMP	DEFAULT NOW()	Detection timestamp

chat_history

Column	Type	Constraint	Description
id	VARCHAR(36)	PRIMARY KEY	UUID
user_id	VARCHAR(36)	FK → users.id	Owning farmer
question	TEXT	NOT NULL	Farmer's question
response	TEXT	NOT NULL	Gemini AI response
date	TIMESTAMP	DEFAULT NOW()	Conversation timestamp

weather_data

Column	Type	Constraint	Description
id	VARCHAR(36)	PRIMARY KEY	UUID
temperature	DECIMAL(5,2)	NOT NULL	Temperature in °C
humidity	DECIMAL(5,2)	NOT NULL	Humidity percentage
rainfall	DECIMAL(6,2)	NULL	Rainfall in mm
flood_risk	VARCHAR(20)	NULL	none / low / medium / high
date	TIMESTAMP	DEFAULT NOW()	Snapshot timestamp

market_prices

Column	Type	Constraint	Description
id	VARCHAR(36)	PRIMARY KEY	UUID
crop_name	VARCHAR(100)	NOT NULL	Name of the crop
price	DECIMAL(8,2)	NOT NULL	Price per unit
unit	VARCHAR(20)	NOT NULL	kg / ton / piece
region	VARCHAR(100)	NULL	Region in Bangladesh
date	TIMESTAMP	DEFAULT NOW()	Price record timestamp

fertilizer_recommendations

Column	Type	Constraint	Description
id	VARCHAR(36)	PRIMARY KEY	UUID
user_id	VARCHAR(36)	FK → users.id	Owning farmer
disease_id	VARCHAR(36)	FK → disease_history.id	Linked disease record (nullable)
crop_name	VARCHAR(100)	NOT NULL	Target crop
suggestion	TEXT	NOT NULL	AI fertilizer advice with dosage
date	TIMESTAMP	DEFAULT NOW()	Recommendation timestamp

ii. Relationships

Relationship	Type	Description
users → disease_history	One-to-Many (1:N)	One farmer can have many disease detection records
users → chat_history	One-to-Many (1:N)	One farmer can have many chatbot conversations
users → fertilizer_recommendations	One-to-Many (1:N)	One farmer receives many fertilizer suggestions
disease_history → fertilizer_rec.	One-to-Zero/One	A disease record may trigger one fertilizer recommendation
weather_data (standalone)	Independent	Populated from Weather API; no user FK
market_prices (standalone)	Independent	Populated from Market API; no user FK

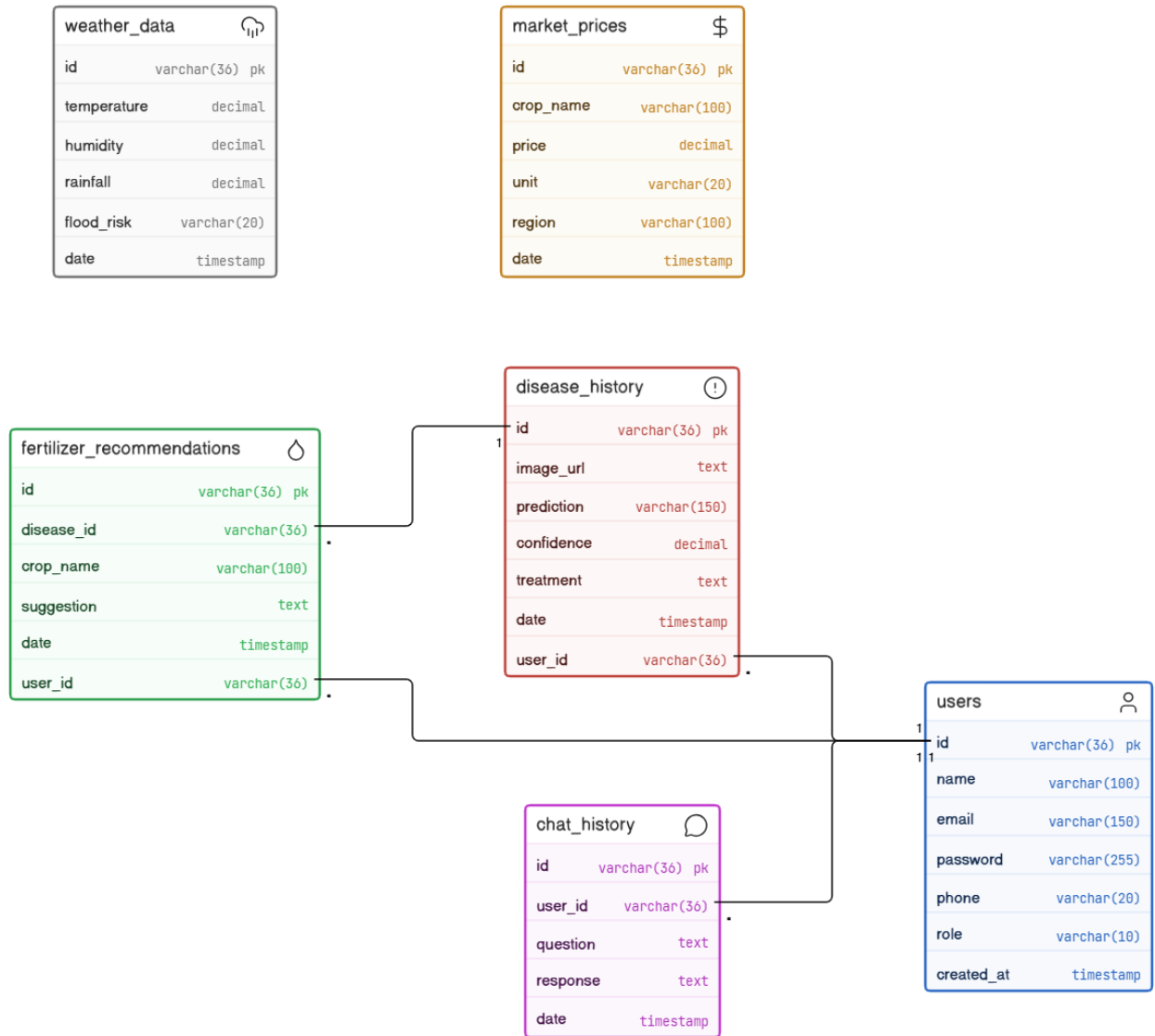


Figure 1: ER Diagram